# Davide Gerosa | Curriculum Vitae

davide.gerosa@unimib.it • www.davidegerosa.com • April 7, 2026

*Relativistic astrophysicist, studying the impact of Einstein's general relativity on the astrophysical world. Research interests include gravitational-wave astronomy, black-hole binary dynamics, Bayesian statistics, and machine-learning applications.*

## Contact

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**Website**: www.davidegerosa.com

## Academic positions

**Università degli Studi di Milano-Bicocca** **Milan, Italy**
*Associate Professor (Professore Associato), Department of Physics* *2021 - current*

- Group leader of ERC project "*GWmining*"; undergraduate and graduate teaching; PhD, MSc and BSc thesis advisor; postdoc advisor; Erasmus program coordinator.
- Research appointment ("incaricato di ricerca") at INFN (National Institute for Nuclear Physics); local PI of the TEONGRAV INFN national initiative.
- Member of the Astronomy Working Group; advisory team of the European Space Agency (ESA).

**University of Birmingham** **Birmingham, UK**
*Assistant Professor (Lecturer), Institute for Gravitational Wave Astronomy* *2019-2021*

- Research-focused appointment; graduate and undergraduate teaching; MSc and PhD student mentoring; postdoc line manager; BSc student tutoring.

**California Institute of Technology** **Pasadena CA, USA**
*Postdoctoral scholar, NASA Einstein Fellow* *2016-2019*

- Now part of the NASA Hubble Fellowship program, Einstein Fellowships are prestigious awards in theoretical astrophysics consisting of personal research grant at selected US institution.

## Education

**University of Cambridge** **Cambridge, UK**
*Ph.D., Department of Applied Mathematics and Theoretical Physics* *2013-2016*

- *Support*: Isaac Newton Studentship; STFC Ph.D. Studentship. The Isaac Newton Studentship is the only astronomy-specific Ph.D. scholarship of the University of Cambridge.
- *Supervisor*: U. Sperhake. Thesis resulted in 14 publications.

**Università degli Studi di Milano** **Milan, Italy**
*Master's degree in Astrophysics* *2010-2013*

- *Final degree grade*: 110/110 with distinction ("cum laude").
- *Average class grade*: 30.00/30 with 7/12 distinctions ("cum laude").
- *Thesis advisors*: E. Berti, G. Lodato. Thesis resulted in 2 publications.

**Università degli Studi di Milano** **Milan, Italy**
*Bachelor's degree in Physics* *2007-2010*

- *Final degree grade*: 110/110 with distinction ("cum laude").
- *Average class grade*: 29.84/30 with 18/26 distinctions ("cum laude").

**Liceo Scientifico Statale Paolo Frisi** **Monza, Italy**
*Scientific High School degree* *2002-2007*

- *Final grade*: 100/100.

**Internships:**

- Institut Astrophysique de Paris, Paris, France. *Jul 2015*
- University of Mississippi, Oxford MS, USA. *Aug-Dec 2012*
- Caltech, Pasadena CA, USA. LIGO Summer Undergraduate Research Fellow (SURF). *Jun-Aug 2012*

## Metrics

**Publications**: **111** papers published in major peer-reviewed journals, **8** papers in submission stage, **14** other publications (white papers, proceedings, etc.) (out of which **33** first-authored papers and **8** papers covered by press releases).

Summary metrics reported using ADS and InSpire excluding [including] long-authorlist papers:
**Total number of citations**: >6600 [>11000] — **h-index**: 44 [48].
**Web links to list services**: ADS; InSpire; arXiv; ORCID.
**Full list of publications** available below and at `www.davidegerosa.com/pub`.

**Presentations**: **84** talks at conferences, **47** talks at department seminars, **8** posters at conferences, (out of which **86** invited presentations), **9** lectures at PhD schools, **18** outreach talks.

**Full list of presentations** available below and at `www.davidegerosa.com/talks`.

## Grants

For collaborative grants, the reported amount refers to the portion awarded to my unit/group.

**Italian-French University (UIF/UFI) - Chapter 3 Grant**. PI, unit leader, 78k EUR, No. 2025-C3-386. *2025-2029*
*"Populations of compact objects for next-generation gravitational-wave detectors"*

**Ministero Università e Ricerca (MUR) – PRIN Grant**. Co-I, unit leader, 85k EUR. No. 2022-Z9X4XS *2023-2025*
*"Gravitational-wave astronomy as a mature field: selection biases and environmental effects"*

**European Research Council (ERC) – Starting Grant**. PI, 1.5M EUR, No. 945155–GWmining. *2021-2026*
*"Gravitational-wave data mining"*

**Cariplo Foundation – Excellence Attractiveness Grant**. PI, 330k EUR, No. 2021-0555. *2021-2026*
*"Deep into the relativistic two-body problem"*

**Leverhulme Trust – Research Project Grant**. PI, 190k GBP, No. RPG-2019-350. *2020-2023*
*"Black-hole spins and accretion discs with gravitational waves from space"*

**Royal Society – Research Grant**. PI, 20k GBP, No. RGS-R2-202004. *2020-2021*
*"The supermassive black-hole binary puzzle: putting the pieces together"*

## Awards

**Career prizes:**

- PRD Reviewer Excellence Award, American Physical Society. *2025*
- Frontiers of Science Award (Physics), International Congress of Basic Science, Beijing, China. *2025*
- SIGRAV Prize for Young Researchers, Italian Society of General Relativity and Gravitational Physics. *2021*
- Research Prize of the Italian Minister of Cultural Heritage, Lincei National Academy. *2020*
- IUPAP GRG Young Scientist Prize, International Society for General Relativity and Gravitation. *2020*
- Giulio Rampa PhD Thesis Prize, Italian Society of General Relativity and Gravitational Physics. *2018*
- Stefano Braccini PhD Thesis Prize, Gravitational Wave International Committee (GWIC). *2017*

**Personal scholarships:**

- Einstein Postdoctoral Prize Fellowship, NASA. *2016-2019*
- Isaac Newton PhD Studentship, DAMTP & IoA, University of Cambridge. *2013-2016*
- STFC PhD Studentship, UK Science & Technology Facilities Council. *2013-2016*
- Rouse Ball Studentship in Mathematics, Trinity College, University of Cambridge. *2015*
- LIGO Summer Undergraduate Research Fellowship (SURF), California Institute of Technology. *2012*

- Undergraduate Studentship, Physics Department, University of Milan. *2007-2008*

**Other funding:**

- Fondo Ateneo Quota Competitiva (internal competitive grant at Milano-Bicocca), 15k EUR. *2025*
- MUR Young Researchers Fellowship (Italy), supervisor for candidate A. Toubiana, 250k EUR. *2025*
- **Marie Skłodowska-Curie Fellowship** (EU), supervisor for candidate Z. Roupas, 188k EUR. *2024*
- **Marie Skłodowska-Curie Fellowship** (EU), supervisor for candidate A. Renzini, 188k EUR. *2023*
- Fondo Ateneo Quota Competitiva (internal competitive grant at Milano-Bicocca), 25k EUR. *2023*
- CINECA ISCRA Type-B project, 1.2M CPUh at the Italian National HPC center. *2022*
- Stephen Hawking Fellowship (UK), supervisor for candidate S. Baghwat, 460k GBP. *2022*
- HPC Baskerville pilot project, early GPU access at UK Tier 2 cluster. *2021*
- Undergraduate summer student support, Royal Astronomical Society (UK), 1k GBP. *2020*
- HPC Midlands+ substantial project, 3M CPUh at UK Tier 2 cluster. *2020*
- NSF Grant PHY-1759835, National Science Foundation (USA), 10k USD. *2018*
- Travel Grant, Royal Astronomical Society, UK. *2016*
- Smith-Rayleigh-Knight Essay Prize, Faculty of Mathematics, University of Cambridge. *2015*

## Research group

**Postdoc fellowships**: **4** so far (of which **2** currently in my group).
**Postdocs (hired)**: **11** so far (of which **5** currently in my group).
**PhD students**: **10** so far (of which **5** currently in my group).
**MSc students**: **35** so far (of which **7** currently in my group).
**BSc students**: **50** so far (of which **4** currently in my group).

Current group members marked with *. More information at `www.davidegerosa.com/group`

**Postdoc external fellowship sponsor:**

- *Alexandre Toubiana* (Milano-Bicocca, MUR Young Researchers Fellow).* *2025-current*
- *Zacharias Roupas* (Milano-Bicocca, ERC Marie Skłodowska-Curie Fellow).* *2024-current*
- *Arianna Renzini* (Milano-Bicocca, ERC Marie Skłodowska-Curie Fellow). *2023-2025*
- *Swetha Baghwat* (Birmingham, UKRI Stephen Hawking Fellow). *2022-2025*

**Hired postdocs:**

- *Caroline Owen* (Milano-Bicocca).* *2024-current*
- *Tristan Bruel* (Milano-Bicocca).* *2024-current*
- *Rodrigo Tenorio* (Milano-Bicocca).* *2024-current*
- *Ssohrab Borhanian* (Milano-Bicocca).* *2023-current*
- *Nicholas Loutrel* (Milano-Bicocca).* *2023-current*
- *Philippa Cole* (Milano-Bicocca). *2023-2026*
- *Costantino Pacilio* (Milano-Bicocca). *2022-2025*
- *Alexandre Toubiana* (Milano-Bicocca). *2024-2025*
- *Michele Mancarella* (Milano-Bicocca). *2022-2024*
- *Nathan Steinle* (Birmingham). *2021-2023*
- *Nicola Giacobbo* (Birmingham). *2020-2021*

**PhD student supervisor:**

- *Giulia Cuomo* (Milano-Bicocca).* *2025-current*
- *Matilde Garcia* (Milano-Bicocca).* *2025-current*
- *Chiara Anselmo* (Milano-Bicocca).* *2024-current*
- *Federico De Santi* (Milano-Bicocca).* *2024-current*
- *Matteo Boschini* (Milano-Bicocca).* *2023-current*
- *Alice Spadaro* (Milano-Bicocca). *2022-2025*
- *Giulia Fumagalli* (Milano-Bicocca). *2022-2025*
- *Viola De Renzis* (Milano-Bicocca). *2021-2024*
- *Daria Gangardt* (Birmingham). *2020-2024*
- *Matthew Mould* (Birmingham). *2019-2023*

**MSc student mentoring:**
*G. Foroni* (Milano-Bicocca and Columbia, MSc thesis, 2026)* — *R. Giugni* (Milano-Bicocca and Roma-Sapienza, MSc thesis, 2026)* — *L. Merlo* (Milano-Bicocca, MSc thesis, 2026)* — *S. Dossena* (Milano-Bicocca, MSc thesis, 2026)* — *M. Bianchi* (Milano-Bicocca, MSc thesis, 2026)* — *G. Monti* (Milano-Bicocca and GSSI L'Aquila, MSc thesis, 2026)* — *S. Caslini* (Milano-Bicocca, MSc thesis, 2026)* — *P. Zeduri* (Milano-Bicocca and Aix-Marseille, MSc thesis, 2026) — *A. Corelli* (Milano-Bicocca and Nottingham, MSc thesis, 2026) — *F. Leto di Priolo* (Milano-Bicocca and ESO Garching, MSc thesis, 2026) — *O. Rossi* (Milano-Bicocca, MSc thesis, 2026) — *G. Conti* (Milano-Bicocca, MSc thesis, 2026) — *M. Gerini* (Milano-Bicocca, MSc thesis, 2026) — *E. Sottocorno* (Milano-Bicocca, MSc thesis, 2025) — *S. Restuccia* (Milano-Bicocca, MSc thesis, 2025) — *L. Toti* (Milano-Bicocca, MSc thesis, 2025) — *O. Pietrosanti* (Milano-Bicocca, MSc thesis, 2025) — *N. Grillo* (Milano-Bicocca, MSc thesis, 2025) — *G. Giarda* (Milano-Bicocca, MSc thesis, 2025) — *F. Tettoni* (Milano-Bicocca, MSc thesis, 2024) — *C. Fabbri* (Milano-Bicocca, MSc thesis, 2024) — *A. Pedrotti* (Milano-Bicocca, MSc thesis, 2024) — *F. Nobili* (Milano-Bicocca and Birmingham, MSc thesis, 2023) — *A. Santini* (Milano-Bicocca and Johns Hopkins, MSc thesis, 2023) — *S. Piscitelli* (Milano-Bicocca and Milano-Statale, MSc thesis, 2023) — *M. Boschini* (Milano-Bicocca and Max Planck AEI, MSc thesis, 2023) — *G. Cavallotto* (Milano-Bicocca, MSc thesis, 2022) — *A. Spadaro* (Milano-Bicocca, MSc thesis, 2022) — *A. Carzaniga* (Milano-Bicocca, MSc thesis, 2022) — *A. Geminardi* (Milano-Bicocca, MSc thesis, 2022) — *M. Dabrowny* (Birmingham, Year 4 project, 2021) — *B. Basset* (Birmingham and Lyon, Final-year internship, 2020) — *J. Chan* (Birmingham, Year 4 project, 2020) — *A. Aziz* (Birmingham, Year 4 project, 2020) — *R. Barbieri* (Caltech and Pavia, MSc thesis, 2018).

**BSc student projects:**
*F. Massacesi* (Milano-Bicocca, BSc thesis, 2026)* — *N. Toselli* (Milano-Bicocca, BSc thesis, 2026)* — *A. Zappietro* (Milano-Bicocca and Pavia, BSc thesis, 2026)* — *P. Basta* (Milano-Bicocca, BSc thesis, 2026)* — *S. Abeni* (Milano-Bicocca, BSc thesis, 2026) — *L. Lecci* (Milano-Bicocca, BSc thesis, 2025) — *F. Quattrini* (Milano-Bicocca, BSc thesis, 2025) — *A. Pedone* (Milano-Bicocca, BSc thesis, 2025) — *S. Scarlett* (Milano-Bicocca and Boston University, IREU summer project, 2025) — *A. Malfasi* (Milano-Bicocca, BSc thesis, 2025) — *R. Giugni* (Milano-Bicocca, BSc thesis, 2024) — *G. Foroni* (Milano-Bicocca, BSc thesis, 2024) — *M. Vergani* (Milano-Bicocca, BSc thesis, 2024) — *L. Tassoni* (Milano-Bicocca, BSc thesis, 2024) — *F. Rattegni* (Milano-Bicocca, BSc thesis, 2024) — *M. Pagani* (Milano-Bicocca, BSc thesis, 2024) — *W. Toscani* (Milano-Bicocca, BSc thesis, 2024) — *A. Bailey* (Milano-Bicocca and Duke, IREU summer project, 2024) — *A. Crespi* (Milano-Bicocca, BSc thesis, 2024) — *A. Amigoni* (Milano-Bicocca, BSc thesis, 2024) — *A. Palladino* (Milano-Bicocca, BSc thesis, 2024) — *L. Merlo* (Milano-Bicocca, BSc thesis, 2024) — *S. Caslini* (Milano-Bicocca, BSc thesis, 2023) — *M. Falcone* (Milano-Bicocca, BSc thesis, 2023) — *M. Bianchi* (Milano-Bicocca, BSc thesis, 2023) — *M. Gerini* (Milano-Bicocca, BSc thesis, 2023) — *F. Ravelli* (Milano-Bicocca, BSc thesis, 2023) — *S. Sferlazzo* (Milano-Bicocca, BSc thesis, 2023) — *M. Bellotti* (Milano-Bicocca, BSc thesis, 2023) — *R. Bosoni De Martini* (Milano-Bicocca, BSc thesis, 2023) — *L. Carbone* (Milano-Bicocca, BSc thesis, 2023) — *H. Blake* (Milano-Bicocca and Ohio State, IREU summer project, 2023) — *L. Toti* (Milano-Bicocca, BSc thesis, 2023) — *S. Restuccia* (Milano-Bicocca, BSc thesis, 2023) — *D. Chirico* (Milano-Bicocca, BSc thesis, 2023) — *M. Muriano* (Milano-Bicocca, BSc thesis, 2022) — *L. Zanga* (Milano-Bicocca, BSc thesis, 2022) — *O. Rossi* (Milano-Bicocca, BSc thesis, 2022) — *S. Neogi* (Birmingham and IISER Pune, Summer project, 2022) — *D. Padilla Monroy* (Milano-Bicocca and Florida International, IREU summer project, 2022) — *S. Al Humaikani* (Birmingham and ENSTA Paris, summer project, 2022) — *N. Sivrioglu* (Milano-Bicocca and Grinell College, Summer project, 2022) — *C. Fabbri* (Milano-Bicocca, BSc thesis, 2022) — *M. Vogel* (Birmingham and Missouri State, IREU summer project, 2021) — *D. Gangardt* (Birmingham, Summer project, 2019) — *L. Reali* (Birmingham and Milano-Statale, BSc thesis, 2019) — *A. Lima* (Caltech and Bowdoin College, SURF summer project, 2018) — *K. Chamberlain* (Caltech and Montana State, SURF summer project, 2017) — *R. Barbieri* (Cambridge and Pavia, Summer project, 2016) — *J. Vosmera* (Cambridge, Summer project, 2015).

## Teaching

**Taught classes:**

- *Machine learning applied to physical systems.* *2024-current*
  Artificial Intelligence BSc, Pavia+Milano+Milano-Bicocca joint degree.
  `dgerosa.github.io/machinelearning4physics`
- *Scientific computing with python.* *2023-current*

Physics PhD, Milano-Bicocca.
`dgerosa.github.io/scientificcomputing`

- *Astrostatistics and machine learning*. *2022-current*
  Astrophysics and Space Science MSc, Milano-Bicocca.
  `dgerosa.github.io/astrostatistics`
- *Classical physics*. *2021-2023*
  Computer Science BSc, Milano-Bicocca.
- *Year 1 Astrolab*. *2020-2021*
  Physics BSc, Birmingham.
- *Year 1 student tutoring*. *2019-2021*
  Physics BSc, Birmingham.
- *Black holes and gravitational waves*. *2019-2020*
  Midlands Physics Alliance Graduate School (MPAGS) PhD, Birmingham.
- *Part III General relativity* (teaching assistant). *2015-2016*
  Mathematics MSc, Cambridge.
- *Part II General relativity* (teaching assistant). *2014-2016*
  Natural Sciences BSc, Cambridge.

**Teaching qualifications:**

- UK Postgraduate Certificate in Higher Education (PGCHE). Graduated with distinction. *2019-2021*

## Service

**Journal referee**

- Nature Astronomy,
- Nature Communications,
- Science,
- Science Advances,
- Physical Review Letters,
- Physical Review D,
- The Astrophysical Journal Letters,
- The Astrophysical Journal,
- Monthly Notices of the Royal Astronomical Society,
- Classical and Quantum Gravity,
- Astronomy & Astrophysics
- General Relativity and Gravitation,
- Reports on Progress in Physics,
- European Physical Journal Plus,
- European Physical Journal C,
- Europhysics Letters,
- Physics Letters A,
- Scientific Reports,
- Annals of Physics
- Astrophysics and Space Science,
- Journal of Open Source Software,
- SoftwareX,
- Taylor and Francis Press (textbooks).

I refereed papers written by the LIGO/Virgo/KAGRA Collaboration, the NanoGRAV Collaboration, and the SXS Collaboration.

**Journal editor**

- Classical and Quantum Gravity, Focus Issue *"Numerical Relativity Beyond General Relativity"*. *2019*

**Grant proposal reviewer**. Panel memberships marked with *.

- ERC, Starting Grants, European Union;
- ERC, Advanced Grants, European Union;
- COST Actions, European Union;
- ESA, Voyage 2050 Topical Teams, Europe;*
- NASA, Astrophysics Theory Program (ATP), USA;*
- NASA, Hubble Fellowship Program (NHFP), USA;*
- NASA, Postdoctoral Program (NPP), USA;
- NASA, Graduate Fellowship Program (NESSF/FINESST), USA;
- NSF, CAREER program, USA;
- NSF, Physics Division Grants, USA;*

- NSF, Astronomy Division Grants, USA;*
- STFC, Gravitational Waves Consolidated Grant Panel, UK;*
- STFC, Astronomy Consolidated Grant Panel, UK;
- STFC, Rutherford Fellowships, UK;
- EPSRC, Hawking Fellowships, UK;
- Royal Society, University Research Fellowships, UK;
- STFC, DiRAC HPC cluster, UK;*
- MUR, Montalcini Fellowships, Italy;
- MUR, Valutazione della Qualità della Ricerca (VQR), Italy;
- Cariplo Foundation, Italy;
- Humboldt Research Fellowships, Germany;
- Dim Origines Grants, France;
- NWO, Veni Fellowships, The Netherlands;
- Swiss NSF, Ambizione Fellowships, Switzerland;
- Irish Research Council, Laureate Programme, Ireland;
- Fonds de la Recherche Scientifique, Grants and Fellowship Call, Belgium;
- National Science Centre, Poland;
- Research Grants Council, Hong Kong.

**Hiring board member** for University of Birmingham (UK), University of Milano-Bicocca (Italy), Sapienza University of Rome (Italy), University of Padua (Italy).

**PhD thesis examiner** for SISSA (International School for Advanced Studies; Italy), Imperial College London (UK), Monash University (Australia), University of Geneva (Switzerland), Cardiff University (UK), University of Pisa (Italy), University of Milan (Italy).

**Organized conferences**

- Binary compact objects on eccentric orbits and their gravitational-wave signatures *2026*
  Corfu Summer Institute, Greece. [link]
- GWfreeride: carving the AI gradient in gravitational-wave astronomy *2026*
  Sexten Center for Astrophysics, Italy. [link]
- XXVI SIGRAV conference on general relativity and gravitation *2025*
  University of Milano-Bicocca, Italy. [link]
- Theoretical Horizons in Unraveling Relativity, Astrophysics, and Mergers (THURAM). *2024*
  Gran Sasso Science Institute, Italy. [link]
- Gravitational-wave snowballs, populations, and models *2025*
  Sexten Center for Astrophysics, Italy. [link]
- Challenges and future perspectives in gravitational-wave astronomy: O4 and beyond *2024*
  Lorentz Center, Leiden, the Netherlands. [link]
- Linking Advances in our Understanding of Theoretical Astrophysics and Relativity to Observations *2024*
  (LAUTARO). University of Milano-Bicocca, Italy. [link]
- Big data within science and industry. *2023*
  University of Milano-Bicocca, Italy. [link]
- Gravitational-wave populations: what's next? *2023*
  University of Milano-Bicocca, Italy. [link]
- Gravitational-wave Excellence Alliance Training (GrEAT) PhD school. *2019*
  University of Birmingham, UK. [link]
- Numerical Relativity beyond General Relativity. *2018*
  Benasque Science Center, Benasque, Spain. [link]
- APS 34$^{th}$ Pacific Coast Gravity Meeting. *2018*
  Caltech, USA. [link]
- The migration issue: from protoplanets to supermassive black holes. *2017*
  University of Cambridge, UK. [link]
- Einstein's Legacy: Celebrating 100yrs of General Relativity. *2015*
  Queen Mary University of London, UK.

**Professional recognition and service**

- **Appointed member of the Astronomy Working Group of the European Space Agency.** *2026 - current*
- Local coordinator, TEONGRAV initiative, National Institute for Nuclear Physics (INFN). *2025 - current*
- Erasmus program coordinator, Physics department, University of Milano-Bicocca. *2024 - current*
- European Consortium for Astroparticle Theory (EuCAPT), council member. *2022 - current*
- Italian Habilitation to Full Professorship in Astrophysics (ASN 02/C1). *2022*
- Italian Habilitation to Full Professorship in Theoretical Physics (ASN 02/A2). *2022*
- Italian Habilitation to Associate Professorship in Astrophysics (ASN 02/C1). *2020*
- Italian Habilitation to Associate Professorship in Theoretical Physics (ASN 02/A2). *2018*
- Seminar organizer, AstroBicocca seminar series, Milano-Bicocca. *2021-2024*
- Academic advisory team for the BlueBEAR HPC cluster, Birmingham. *2019-2021*
- Seminar organizer, TAPIR seminar series, Caltech. *2018-2019*
- PhD student representative, Gravitational Physics committee, UK Institute of Physics. *2014-2016*
- Undergraduate student representative, Physics Department, University of Milan. *2008-2012*

**Memberships**

- International Society on General Relativity and Gravitation (ISGRG), life member.
- Italian Society of General Relativity and Gravitational Physics (SIGRAV), honorary life member.
- International Astronomical Union (IAU), Individual member.
- American Physical Society (APS), member.
- Italian Physical Society (SIF), member.
- Italian Center for Supercomputing (ICSC).
- TEONGRAV national initiative (Gravity Theory), National Institute for Nuclear Physics (INFN).
- AI_INFN National Initiative (Artificial Intelligence), National Institute for Nuclear Physics (INFN).
- LISA Consortium, community member.
- Foundational Questions Institute (FQxI), member.

## Skills

**Programming languages**: Python (advanced), Bash, Mathematica, C.

**Other scientific tools**: TensorFlow, pytorch, LIGO lalsuite, LaTex, git, slurm, cloud computing, website dev.

**Languages**: English (fluent), Italian (native).

## Hobbies

Alpinism, rock climbing, skiing. Climbed to the top of Monte Rosa, 2nd highest mountain in the Alps (next dream: the Matterhorn). In love with the US National Parks. Bruce Springsteen fanatic. Too many fantasy & sci-fi books, Tex Willer comics. FC Inter fan. Some running as well. Very poor attempts at photography.

## Full publication list

### Submitted papers:

8. *Accurate and efficient simulation-based inference for massive black-hole binaries with LISA.*
A. Spadaro, J. Gair, **D. Gerosa**, S. R. Green, R. Buscicchio, N. Gupte, R. Tenorio, S. Clyne, M. Purrer, N. Korsakova.
arXiv:2603.20431 [astro-ph.HE].
7. *Variance of gravitational-wave populations.*
A. Corelli, **D. Gerosa**, M. Mould, C. M. Fabbri.
arXiv:2603.00239 [astro-ph.HE].
6. *Inferring the population properties of galactic binaries from LISA's stochastic foreground.*
F. De Santi, A. Santini, A. Toubiana, N. Karnesis, **D. Gerosa**.
arXiv:2602.18560 [astro-ph.HE].
5. *On the exceptionality of exceptional gravitational-wave events.*
R. Tenorio, **D. Gerosa**.
arXiv:2601.02467 [astro-ph.HE].
4. *Black-hole ringdown with templates capturing spin precession: a criticalre-analysis of GW190521.*
C. Anselmo, C. Pacilio, **D. Gerosa**.
arXiv:2512.05193 [gr-qc].
3. *Cosmology with the angular cross-correlation of gravitational-wave and galaxy catalogs: forecasts for next-generation interferometers and the Euclid survey.*
A. Pedrotti, M. Mancarella, J. Bel, **D. Gerosa**.
arXiv:2504.10482 [astro-ph.CO].
2. *A recoiling supermassive black hole in a powerful quasar.*
M. Chiaberge, T. Morishita, M. Boschini, S. Bianchi, A. Capetti, G. Castignani, **D. Gerosa**, M. Konishi, S. Koyama, K. Kushibiki, E. Lambrides, E. T. Meyer, K. Motohara, M. Stiavelli, H. Takahashi, G. R. Tremblay, C. Norman.
arXiv:2501.18730 [astro-ph.GA].
1. *The last three years: multiband gravitational-wave observations of stellar-mass binary black holes.*
A. Klein, G. Pratten, R. Buscicchio, P. Schmidt, C. J. Moore, E. Finch, A. Bonino, L. M. Thomas, N. Williams, **D. Gerosa**, S. McGee, M. Nicholl, A. Vecchio.
arXiv:2204.03423 [gr-qc].

### Papers published in major peer-reviewed journals:

111. *Probing modified gravitational-wave dispersion with bursts from eccentric black-hole binaries.*
N. Loutrel, A. Bailey, **D. Gerosa**.
Classical and Quantum Gravity, in press. arXiv:2509.01614 [gr-qc].
110. *Impact of facility timing and coordination for next-generation gravitational-wave detectors.*
S. Borhanian, A. Renzini, P. S. Cole, C. Pacilio, M. Mancarella, **D. Gerosa**.
Classical and Quantum Gravity, in press. arXiv:2510.11861 [gr-qc].
109. *Comparing astrophysical models to gravitational-wave data in the observable space.*
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**14.** *On the equal-mass limit of precessing black-hole binaries.*
**D. Gerosa**, U. Sperhake, J. Vošmera.
Classical and Quantum Gravity 34 (2017) 064004. arXiv:1612.05263 [gr-qc].

**13.** *Black-hole kicks as new gravitational-wave observables.*
**D. Gerosa**, C. J. Moore.
**Physical Review Letters** 117 (2016) 011101. arXiv:1606.04226 [gr-qc].
• PRL Editors' Suggestion. Covered by press release.

**12.** *PRECESSION: Dynamics of spinning black-hole binaries with python.*
**D. Gerosa**, M. Kesden.
Physical Review D 93 (2016) 124066. arXiv:1605.01067 [astro-ph.HE].
• Open source code.

**11.** *Numerical simulations of stellar collapse in scalar-tensor theories of gravity.*
**D. Gerosa**, U. Sperhake, C. D. Ott.
Classical and Quantum Gravity 33 (2016) 135002. arXiv:1602.06952 [gr-qc].

**10.** *Distinguishing black-hole spin-orbit resonances by their gravitational wave signatures. II: Full parameter estimation.*
D. Trifirò, R. O'Shaughnessy, **D. Gerosa**, E. Berti, M. Kesden, T. Littenberg, U. Sperhake.
Physical Review D 93 (2016) 044071. arXiv:1507.05587 [gr-qc].

**9.** *Precessional instability in binary black holes with aligned spins.*
**D. Gerosa**, M. Kesden, R. O'Shaughnessy, A. Klein, E. Berti, U. Sperhake, D. Trifirò.
**Physical Review Letters** 115 (2015) 141102. arXiv:1506.09116 [gr-qc].
• PRL Editors' Suggestion.

**8.** *Tensor-multi-scalar theories: relativistic stars and 3+1 decomposition.*
M. Horbatsch, H. O. Silva, **D. Gerosa**, P. Pani, E. Berti, L. Gualtieri, U. Sperhake.
Classical and Quantum Gravity 32 (2015) 204001. arXiv:1505.07462 [gr-qc].
• IoP Editor's choice (CQG+, IOPselect).

**7.** *Multi-timescale analysis of phase transitions in precessing black-hole binaries.*
**D. Gerosa**, M. Kesden, U. Sperhake, E. Berti, R. O'Shaughnessy.
Physical Review D 92 (2015) 064016. arXiv:1506.03492 [gr-qc].

**6.** *Spin alignment and differential accretion in merging black hole binaries.*
**D. Gerosa**, B. Veronesi, G. Lodato, G. Rosotti.
Monthly Notices of the Royal Astronomical Society 451 (2015) 3941-3954. arXiv:1503.06807 [astro-ph.GA].

**5.** *Effective potentials and morphological transitions for binary black-hole spin precession.*
M. Kesden, **D. Gerosa**, R. O'Shaughnessy, E. Berti, U. Sperhake.
**Physical Review Letters** 114 (2015) 081103. arXiv:1411.0674 [gr-qc].
• Covered by press release.

**4.** *Missing black holes in brightest cluster galaxies as evidence for the occurrence of superkicks in nature.*
**D. Gerosa**, A. Sesana.
Monthly Notices of the Royal Astronomical Society 446 (2015) 38-55. arXiv:1405.2072 [astro-ph.GA].

3. *Distinguishing black-hole spin-orbit resonances by their gravitational-wave signatures.*
**D. Gerosa**, R. O'Shaughnessy, M. Kesden, E. Berti, U. Spehake.
Physical Review D 89 (2014) 124025. arXiv:1403.7147 [gr-qc].
2. *Resonant-plane locking and spin alignment in stellar-mass black-hole binaries: a diagnostic of compact-binary formation.*
**D. Gerosa**, M. Kesden, E. Berti, R. O'Shaughnessy, U. Sperhake.
Physical Review D 87 (2013) 10, 104028. arXiv:1302.4442 [gr-qc].
1. *Black hole mergers: do gas discs lead to spin alignment?*
G. Lodato, **D. Gerosa**.
Monthly Notices of the Royal Astronomical Society Letters 429 (2013) L30-L34. arXiv:1211.0284 [astro-ph.CO].

## Other publications (white papers, proceedings, etc.):

14. *Coincident morphological transitions in precessing black-hole binaries.*
**D. Gerosa**, G. Foroni, G. Fumagalli, E. Berti.
Proceedings of the International Congress of Basic Science, International Press. arXiv:2508.19735 [gr-qc].
13. *PRECESSION 2.1: black-hole binary spin precession on eccentric orbits.*
G. Fumagalli, **D. Gerosa**, N. Loutrel.
Journal of Physics: Conference Series 3177 (2026) 012117. arXiv:2508.21125 [gr-qc].
12. *Waveform modelling for the Laser Interferometer Space Antenna.*
N. Afshordi, et al. (105 authors incl. **D. Gerosa**).
Living Reviews in Relativity 28 (2025) 9. arXiv:2311.01300 [gr-qc].
11. *QLUSTER: quick clusters of merging binary black holes.*
**D. Gerosa**, M. Mould.
Proceedings of the 57th Rencontres de Moriond. arXiv:2305.04987 [astro-ph.HE].
• Open source code.
10. *Astrophysics with the Laser Interferometer Space Antenna.*
P. Amaro-Seoane, et al. (155 authors incl. **D. Gerosa**).
Living Reviews in Relativity 26 (2022) 2. arXiv:2203.06016 [gr-qc].
9. *New horizons for fundamental physics with LISA.*
K. G. Arun, et al. (141 authors incl. **D. Gerosa**).
Living Reviews in Relativity 25 (2022) 4. arXiv:2205.01597 [gr-qc].
8. *Prospects for fundamental physics with LISA.*
E. Barausse, et al. (320 authors incl. **D. Gerosa**).
General Relativity and Gravitation 52 (2020) 8, 81. arXiv:2001.09793 [gr-qc].
7. *Black holes, gravitational waves and fundamental physics: a roadmap.*
L. Barack, et al. (199 authors incl. **D. Gerosa**).
Classical and Quantum Gravity 36 (2019) 143001. arXiv:1806.05195 [gr-qc].
6. *Reanalysis of LIGO black-hole coalescences with alternative prior assumptions.*
**D. Gerosa**, S. Vitale, C.-J. Haster, K. Chatziioannou, A. Zimmerman.
Proceedings of the International Astronomical Union 338 (2018) 22-28. arXiv:1712.06635 [astro-ph.HE].
5. *Surprises from the spins: astrophysics and relativity with detections of spinning black-hole mergers.*
**D. Gerosa**.
Journal of Physics: Conference Series 957 (2018) 012014. arXiv:1711.10038 [astro-ph.HE].
4. *filltex: Automatic queries to ADS and INSPIRE databases to fill LaTex bibliography.*
**D. Gerosa**, M. Vallisneri.
Journal of Open Source Software 2 (2017) 13.
• Open source code.
3. *Testing general relativity with present and future astrophysical observations.*
E. Berti, et al. (53 authors incl. **D. Gerosa**).
Classical and Quantum Gravity 32 (2015) 243001. Topical Review. arXiv:1501.07274 [gr-qc].
2. *Rival families: waveforms from resonant black-hole binaries as probes of their astrophysical formation history.*
**D. Gerosa**.
Astrophysics and Space Science Proceedings 40 (2015) 137-145.
1. *Spin alignment effects in black hole binaries.*
**D. Gerosa**.
Caltech Undergraduate Research Journal (CURJ) 15:1 (2014) 17-26.

## Full presentation list

Invited talks marked with *.

### Talks at conferences:

**84.** *Are exceptional gravitational-wave events truly exceptional?.*
GWfreeride: Carving the AI gradient in gravitational-wave astronomy, Sexten, Italy, Jan 2026.

**83.*** *When measuring binaries, identifying each object is ambiguous.*
7th Milan Christmas Workshop, Milan, Italy, Dec 2025.

**82.** *The full gravitational-wave population problem (aka: sample them all).*
XXVI SIGRAV Conference 2021, Milano, Italy, Sep 2025.

**81.*** *Spin precession across timescales.*
EOB@Work25, Torino, Italy, Sep 2025.

**80.*** *Stacking gravitational-wave events for testing GR: proceed with care.*
Testing Aspects of General Relativity-IV, Gandhinagar, India, (online), Jul 2025.

**79.** *Which black hole? The labeling uncertainty in gravitational-wave astronomy.*
24th International Conference on General Relativity and Gravitation (GR24) and 16th Edoardo Amaldi Conference, Glasgow, UK, Jul 2025.

**78.*** *Are merging black holes born from stellar collapse or previous mergers? Some years later....*
International Congress of Basic Science, Beijing, China, Jul 2025.

**77.*** *Machine learning in gravitational-wave population inference.*
Scientific Machine Learning for Gravitational Wave Astronomy, Providence RI, USA, Jun 2025.

**76.** *Black holes and gravitational waves are discovered (also) with Python!.*
PyCon Italia 2025, Bologna, Italy, May 2025.

**75.** *We observe compact binaries but are interested in single objects, this is hard.*
Theoretical Horizons in Unraveling Relativity, Astrophysics, and Mergers (THURAM), L'Aquila, Italy, May 2025.

**74.*** *Spin the black circle.*
Stellar black hole formation and detection, Kyoto, Japan, Mar 2025.

**73.** *It's not at all trivial to say which is BH1 and which is BH2.*
Gravitational-wave snowballs, populations, and models, Sexten, Italy, Jan 2025.

**72.*** *36 186: the kick strikes back.*
6th Milan Christmas Workshop, Milan, Italy, Dec 2024.

**71.*** *Merge many times.*
Spanish and Portuguese Relativity Meeting (EREP 2024), Coimbra, Portugal, Jul 2024.

**70.*** *One population fit to rule them all.*
Emerging methods in gravitational-wave population inference, Trieste, Italy, Jun 2024.

**69.** *Gravitational-wave selection effects, the easy way.*
Linking Advances in our Understanding of Theoretical Astrophysics and Relativity to Observations (LAUTARO), Milan, Italy, Apr 2024.

**68.** *Quasar 3C186 as a recoiling massive black hole: still possible but unlikely.*
April APS Meeting, Sacramento CA, USA, Apr 2024.

**67.** *Black-hole generations in AGN disks.*
TEONGRAV meeting, Naples, Italy, Dec 2023.

**66.*** *Correlated black holes from AGN disks.*
5th Milan Christmas Workshop, Milan, Italy, Dec 2023.

**65.*** *The masses and spins of LIGO's black holes are correlated... here is a disk explanation.*
RESCEU-NBIA workshop on gravitational-wave sources, Tokyo, Japan, Dec 2023.

**64.*** *Gravitational waves: figuring out what's missing.*
6th World Laureates Forum, Shanghai, China, Nov 2023.

**63.** *A lonely gravitational wave.*
LISA Astrophysics Working Group Meeting, Milan, Italy, Sep 2023.

**62.** *Eccentricity, spin precession, and gravitational-wave signals that are too short.*
15th Edoardo Amaldi Conference, (online), Jul 2023.

**61.** *When orbital eccentricity and spin precession are kind of the same.*
April APS Meeting, Minneapolis MN, USA, Apr 2023.

**60.*** *The black-hole binary formation-channel problem in gravitational-wave astronomy.*
57th Rencontres de Moriond - Gravitation, La Thuile, Italy, Mar 2023.

**59.*** *You'll never merge alone.*
Unsolved problems in astrophysics and cosmology, Jerusalem, Israel, Dec 2022.

**58.** *Characterization of LIGO/Virgo selection effects with neural networks.*
Machine learning in GW search: g2net next challenges, Pisa, Italy, Sep 2022.

**57.** *Machine learning and the origin of LIGO's black holes.*
31th Texas Symposium on Relativistic Astrophysics, Prague, Czechia, Sep 2022.

**56.*** *Gravitational-wave populations with a pinch of deep learning.*
23nd International Conference on General Relativity and Gravitation (GR23), Beijing, China, (online), Jul 2022.

**55.** *Accretion disks around supermassive black-hole binaries can break.*
EuCAPT workshop: Gravitational-wave probes of black hole environments, Rome, Italy, Jun 2022.

**54.*** *Black-hole binary spin precession: new theoretical predictions and current observational evidence.*
9th KAGRA international workshop, Beijing, China, (online), Jun 2022.

**53.*** *A deep-learning solution to the gravitational-wave population problem.*
NBIA Workshop on black-hole dynamics: from gaseous environments to empty space, Copenhagen, Denmark, Jun 2022.

**52.** *Gravitational-wave populations and black-hole spin time travel.*
April APS Meeting, New York NY, USA, Apr 2022.

**51.*** *I've seen a deviation from General Relativity: do you believe me?.*
4th Milan Christmas Workshop, Milan, Italy, Dec 2021.

**50.*** *Filling the gap with hierarchical black-hole mergers.*
4th World Laureates Forum, Shanghai, China, (online), Nov 2021.

**49.*** *Inspiral, merge, repeat: hierarchical black-hole mergers and their gravitational-wave signatures.*
XXIV SIGRAV Conference 2021, Urbino, Italy, Sep 2021.

**48.** *Two precessing spins and how to find them.*
14th Edoardo Amaldi Conference, Melbourne, Australia, (online), Jul 2021.

**47.** *Waveform systematics and gravitational-wave catalogs.*
2nd European Physical Society Conference on Gravitation: measuring gravity, London, UK, (online), Jul 2021.

**46.** *Probing warped accretion disks with gravitational-wave observations of supermassive black-hole mergers.*
European Astronomical Society Annual Meeting (EAS2021), Leiden, The Netherlands, (online), Jun 2021.

**45.*** *Hierarchical black-hole mergers in the LIGO/Virgo band.*
Gravitational Wave Astrophysics Conference 2021, Hefei, China, (online), Jun 2021.

**44.** *Testing GR? Keep calm and mind the waveform.*
24rd Capra Meeting on Radiation Reaction in General Relativity, Waterloo, Canada, (online), Jun 2021.

**43.** *Warped accretion disks with LISA.*
Distorted astrophysical discs: new insights and future directions, Cambridge, UK, (online), May 2021.

**42.** *Detectability of gravitational-wave signals with neural-network classifiers.*
RAS Specialist Meeting – Machine learning and artificial intelligence applied to astronomy 2, London, UK, (online), May 2021.

**41.*** *Transitioning to research independence in gravitational-wave astronomy (my own view on postdoc and grant applications).*
Gravitational Wave Astrophysics for Early Career Scientists, Leiden, The Netherlands, (online), May 2021.

**40.** *Can a computer learn if LIGO and Virgo will observe gravitational waves?.*
APS April Meeting, (online), Apr 2021.

**39.** *A self-consistent treatment of warped accretion disks surrounding spinning supermassive black holes.*
Science at the Horizon: the Next-Generation Event Horizon Telescope, (online), Feb 2021.

**38.*** *Astrophysical black-hole binaries: the elephant in the room.*
Primordial black holes confront gravitational-wave data, Rome, Italy, (online), Feb 2021.

**37.*** *Black holes, gravitational waves, and the BEARs.*
10th Birmingham Environment for Academic Research (BEAR) Conference, Birmingham, UK, (online), Sep 2020.

**36.** *Warped accretion discs, black-hole spins, and LISA.*
13th LISA Symposium, (online), Sep 2020.

**35.** *Multi-timescale post-Newtonian dynamics of spinning black-hole binaries: a status update.*
23rd Capra Meeting on Radiation Reaction in General Relativity, Austin TX, USA, (online), Jun 2020.

**34.*** *An attempt to navigate the astro job market.*
LISA Early Career Scientists (LECS) training workshop, (online), Jun 2020.

**33.** *Second-generation black holes, the pair-instability mass gap, and the escape speed of stellar clusters.*
30th Texas Symposium on Relativistic Astrophysics, Portsmouth, UK, Dec 2019.

**32.** *Testing the no-hair theorem with multiband black-hole binaries.*
30th Texas Symposium on Relativistic Astrophysics, Portsmouth, UK, Dec 2019.

**31.** *Pick the right one: Bayesian model selection on catalogs of gravitational-wave events.*
22nd International Conference on General Relativity and Gravitation (GR22) and 13th Edoardo Amaldi Conference, Valencia, Spain, Jul 2019.

**30.** *Space-ground multiband detections. How many?.*
22nd International Conference on General Relativity and Gravitation (GR22) and 13th Edoardo Amaldi Conference, Valencia, Spain, Jul 2019.

**29.*** *Black-hole kicks: how and where.*
Taipei Gravitational Wave Group (TGWG) Conference, Taipei, Taiwan, Oct 2018.

**28.*** *Forming binary black holes out of stars.*
Taipei Gravitational Wave Group (TGWG) Conference, Taipei, Taiwan, Oct 2018.

**27.*** *Getting ready: exploit LISA to improve LIGO's tests of General Relativity.*
Einstein Fellows Symposium 2018, Cambridge MA, USA, Oct 2018.

**26.** *Runaways: recoiling black holes and their gravitational-wave signatures.*
COSPAR 2018 42nd assembly, Pasadena CA, USA, Jul 2018.

**25.** *An unprecedented opportunity: black-hole spectroscopy with LISA forewarnings and LIGO optimizations.*
12th LISA Symposium, Chicago IL, USA, Jul 2018.

**24.*** *What do LIGO's black holes remember?.*
April APS Meeting, Columbus OH, USA, Apr 2018.

**23.** *Modeling black hole kicks with waveform approximants.*
34rd Pacific Coast Gravity Meeting, Pasadena CA, USA, Mar 2018.

**22.** *Black holes from other black holes?.*
Graviy@Malta 2018, Valletta, Malta, Jan 2018.

**21.*** *The gravitational-wave astronomy revolution.*
2nd Milan Christmas Workshop, Milan, Italy, Dec 2017.

**20.** *Reanalysis of LIGO black-hole coalescences with alternative prior assumptions.*
IAU Symposium 338, Baton Rouge LA, USA, Oct 2017.
- Proceedings published by Cambridge University Press (2018).

**19.*** *Careful with the priors: a reanalysis of LIGO black-hole coalescences.*
Einstein Fellows Symposium 2017, Cambridge MA, USA, Oct 2017.

**18.** *Empty galaxies to constrain black-hole superkicks.*
3rd Swinburne-Caltech Galaxy Workshop, Pasadena CA, USA, Sept 2017.

**17.*** *Surprises from the spins: astrophysics and relativity with detections of spinning black-hole mergers.*
12th Edoardo Amaldi Conference (plenary talk), Pasadena CA, USA, Jul 2017.
- Proceedings published by IoP (2018).

**16.** *Core collapse and compact-object formation to test General Relativity.*
LIGO Core Collapse Supernova Workshop, Pasadena CA, USA, Mar 2017.

**15.** *The curious and simple limit of equal-mass precessing black-hole binaries.*
33rd Pacific Coast Gravity Meeting, Santa Barbara CA, USA, Mar 2017.

**14.*** *Spins remember: spin signatures of astrophysical black hole formation mechanisms.*
Strong Gravity and Binary Dynamics with Gravitational Wave Observations, Oxford MS, USA, Feb 2017.

**13.** *The kick is in the waveform: detection of black-hole recoils.*
The Dawning Era of Gravitational-Wave Astrophysics, Aspen CO, USA, Feb 2017.

**12.** *Kicked waveforms: prospects for direct detection of black hole recoils.*
"April" APS Meeting, Washington DC, USA, Jan 2017.

**11.*** *Getting the most out of gravitational-wave observations: kicks and spin precession.*
Einstein Fellows Symposium 2016, Cambridge MA, USA, Oct 2016.

**10.** *Averaging the average: multi-timescale analysis of precessing black-hole binaries.*
21st International Conference on General Relativity and Gravitation (GR21), New York NY, USA, Jul 2016.

**9.** *Differential accretion means differential alignment.*
BritGrav 16, Nottingham, UK, Apr 2016.

**8.** *A new instability to black-hole spin precession.*
28th Texas Symposium on Relativistic Astrophysics, Geneva, Switzerland, Dec 2015.

**7.** *Binary black-hole spin precession: a tale of three timescales.*
One Hundred Years of Strong Gravity, Lisbon, Portugal, Jun 2015.

6. *Giant and empty: black-hole occupation fraction in brightest cluster galaxies.*
BritGrav 15, Birmingham, UK, Apr 2015.

5.* *Not so fast: gas-driven spin alignment in merging black-hole binaries.*
Milan Christmas Workshop, Milan, Italy, Milan, Italy, Dec 2014.

4. *Missing black holes in brightest cluster galaxies as evidence for the occurrence of superkicks.*
99 years of Black Holes, Potsdam, Germany, May 2014.

3. *Rival families: waveforms from resonant black-hole binaries as probes of their astrophysical formation history.*
3rd Session of the Sant Cugat Forum on Astrophysics, Sant Cugat, Spain, Apr 2014.
• Proceedings published by Springer (2015).

2. *Gravitational-wave signals from stellar-mass black-hole binaries in resonant configurations.*
BritGrav 14, Cambridge, UK, Mar 2014.

1. *Spin alignment effects in stellar mass black hole binaries.*
22nd Midwest Relativity Meeting, Chicago IL, USA, Sep 2012.

## Talks at department seminars:

47.* *Merging binary black holes, the first ten years.*
Physics colloquium, Instituto Superior Técnico, Lisbon, Portugal, Feb 2026.

46.* *What gravitational waves can(not) reveal about merging black holes.*
CPT seminars, Centre de Physique Théorique, Université d'Aix-Marseille, Marseille, France, Oct 2025.

45.* *Challenges and opportunities in gravitational-wave population inference.*
Astrophysics+Theory seminars, Imperial College, London, UK, Sep 2025.

44.* *Gravitational-wave populations: deeper statistics for deeper astrophysics.*
Joint Israeli seminar series on gravitational physics, Israel, (online), Jun 2025.

43.* *The case of quasar 3C 186 as kicked black hole.*
Astronomy seminars, Cardiff University, Cardiff, UK, Jul 2024.

42.* *Black-hole binaries, formation channels, and repeated mergers.*
Gravitational-wave seminars, Università di Pisa, Pisa, Italy, May 2024.

41.* *Deep learning and Bayesian stats to uncover the origin of merging black holes.*
INAF/OAPD seminars, Università degli Studi di Padova, Padua, Italy, Apr 2023.

40.* *Deep-learning emulators for gravitational-wave population fits.*
Theoretical Astrophysics seminars, University of Zurich, Zurich, Switzerland, Feb 2023.

39.* *Comparing gravitational-wave data and stellar-physics predictions with deep learning.*
Theoretical Physics colloquium, University of Cambridge, Cambridge, UK, Feb 2023.

38.* *Gravitational-wave data exploitation with deep learning.*
Astrophysics seminars, Università degli Studi di Milano, Milan, Italy, Jan 2023.

37.* *Some of the LIGO events might have merged a few times already.*
Gr@v seminars, Universidade de Aveiro, Aveiro, Portugal, (online), Nov 2021.

36.* *Entering the data-driven regime of gravitational-wave astronomy.*
Physics Department colloquium, Università degli Studi di Milano-Bicocca, Milan, Italy, (online), Oct 2020.

35.* *Waiting for the next black-hole surprise, from the upper mass gap to GW190412.*
Gravity seminars, University of Mississippi, Oxford MS, USA, (online), Jun 2020.

34.* *Keep the black holes merging: cluster escape speed and the curious case of GW190412.*
Astronomy seminars, Cardiff University, Cardiff, UK, (online), Jul 2020.

33.* *Growing up: the next generation of black holes moves out of the cluster.*
Black Hole Initiative Colloquium, Harvard University, Cambridge MA, USA, (online), Mar 2020.

32.* *Multiple generations of black-hole mergers to pinpoint their astrophysical origin.*
Gravity seminars, University of Southampton, Southampton, UK, Feb 2020.

31.* *Assembling multiple generations of black-hole mergers in the LIGO/Virgo band.*
Theoretical Astrophysics seminars, University of Leicester, Leicester, UK, Feb 2020.

30.* *Predicting LIGO's black holes with LISA.*
AstroParticle seminars, Institute for Fundamental Physics of the Universe, Trieste, Italy, Nov 2019.

29.* *Multiband gravitational waves: event rates, detectability, and tests of gravity.*
Particle Cosmology seminars, University of Nottingham, Nottingham, UK, Nov 2019.

28.* *Understanding birth and growth of binary black holes with gravitational waves.*
Physics Colloquium, California State University Los Angeles, Los Angeles CA, USA, Feb 2019.

27.* *The (astro)physics of black-hole mergers and their gravitational-wave emission.*
Physics and Astronomy Colloquium, California State University Northridge, Los Angeles CA, USA, Jan 2019.

**26.*** *Model selection for large catalogs of gravitational-wave events.*
Cosmology seminars, Johns Hopkins University, Baltimore MD, USA, Nov 2018.

**25.*** *How to kick black holes out of their galaxies.*
Strong Gravity Seminars, Perimeter Institute for Theoretical Physics, Waterloo, Canada, Sep 2018.

**24.*** *(Astro)physical consequences of black-hole recoils.*
CITA Seminars, Canadian Institute for Theoretical Astrophysics, Toronto, Canada, Sep 2018.

**23.*** *Black-hole spins and the astrophysics of LIGO's sources.*
AEI seminars, Albert Einstein Institute, Hannover, Germany, Apr 2018.

**22.*** *Dynamics of spinning black-hole binaries as a tool to uncover their formation pathway.*
AEI seminars, Albert Einstein Institute, Potsdam, Germany, Mar 2018.

**21.*** *Runaways: phenomenology and detectability of black-hole recoils.*
Theoretical Astrophysics seminars, University of Florida, Gainsville FL, USA, Mar 2018.

**20.*** *Testing relativity with past and future supernova explosions.*
Physics colloquia, Montana State University, Bozeman MT, USA, Feb 2018.

**19.*** *More astrophysics out of the first gravitational wave detections.*
GSSI seminars, Gran Sasso Science Institute, L'Aquila, Italy, Jan 2018.

**18.*** *Where do binary black holes come from? How do we find out?.*
Theory seminars, Sapienza Università di Roma, , Rome, Italy, Jan 2018.

**17.*** *Monopole radiation, hyper scalarization and inverse chirps: the promise of testing gravity with stellar collapse.*
Steward Observatory lunch seminars, University of Arizona, Tucson AZ, USA, Nov 2017.

**16.*** *Black-hole binary spin precession: from relativity to astronomy.*
GRITTS seminars, Massachusetts Institute of Technology LIGO Lab, Cambridge MA, USA, Oct 2017.

**15.*** *Constraining the astrophysics behind the first LIGO detections.*
Physics colloquia, University of Texas at Dallas, Richardson TX, USA, Sep 2017.

**14.*** *Binary black hole astrophysics with the first gravitational-wave events.*
Theory group seminars, Universitat de Barcelona, Barcelona, Spain, May 2017.

**13.*** *Astrophysics with the first gravitational-wave events: from supernova asymmetries to multiple black-hole generations.*
GR seminars, Department of Applied Mathematics and Theoretical Physics, Cambridge, UK, May 2017.

**12.*** *Black-hole binaries on the road to merger.*
Oskar Klein Center colloquia, Stockholms Universitet, Stockholm, Sweden, Jan 2017.

**11.*** *A tale of astronomy and relativity: formation and evolution of black-hole binaries.*
Department seminars, Università degli Studi di Milano, Milan, Italy, Apr 2016.

**10.*** *A new paradigm for black-hole spin precession.*
Astrophysics seminars, University of Birmingham, Birmingham, UK, Feb 2016.

**9.*** *New insights on binary black-hole spin precession.*
Relativity division seminars, Albert Einstein Institute, Potsdam, Germany, Feb 2016.

**8.*** *Spontaneous scalarization: a promising avenue for gravitational-wave astronomy.*
GReCO seminars, Institut d'Astrophysique de Paris, Paris, France, Jan 2016.

**7.*** *Innovative multi-timescale approach to binary black holes.*
Theoretical Astrophysics seminars, Università degli Studi di Milano, Milan, Italy, Dec 2015.

**6.*** *Stellar collapse in scalar-tensor theories of gravity: prospects for gravitational-wave astronomy.*
Gravity and Particles seminar, University of Nottingham, Nottingham, UK, Dec 2015.

**5.*** *Core collapse and relativistic stars in scalar-tensor theories of gravity.*
Gravity seminars, STAG Research Center, University of Southampton, Southampton, UK, Oct 2015.

**4.*** *Binary black-hole spin alignment: gas-driven and relativistic inspiral.*
Wednesday seminars, Institute of Astronomy, University of Cambridge, Cambridge, UK, Jun 2015.

**3.*** *Spin-orbit resonances: unveiling black-hole binary dynamics on both stellar-mass and supermassive scale.*
Gravitational Physics seminars, Cardiff University, Cardiff, UK, May 2014.

**2.*** *Spin-orbit resonances: unveiling black-hole binary dynamics on both stellar-mass and supermassive scale.*
GR seminars, Department Applied Mathematics and Theoretical Physics, University of Cambridge, Cambridge, UK, May 2014.

**1.*** *Spin alignment and resonant plane in stellar mass black hole binaries.*
Lunch talks, Leiden Observatory, Leiden, The Netherlands, Jan 2012.

## Lectures at PhD schools:

**9.*** *Gravitational waves from compact binaries.*
Theoretical Aspects of Astroparticle Physics, Cosmology and Gravitation (6h class), Florence, Italy, Mar 2026.

**8.*** *Bayesian inference and stochastic sampling in (astro)physics.*
Visitor program mini-course, Instituto Superior Técnico, Lisbon, Portugal, Feb 2026.

**7.*** *PhDs, postdocs, faculties, and all that.*
AstroCareer Day @ University of Milano-Bicocca, Milan, Italy, Dec 2025.

**6.*** *Tips on academic job applications for early career scientists.*
Gravitational-wave early-career scientists (GWECS) online talks, (online), Nov 2024.

**5.*** *My thoughts on the academic job market.*
IPTA Student Week 2024, Milan, Italy, Jun 2024.

**4.*** *Stellar-mass black-hole binary formation channels.*
Kavli-Villum Summer School on Gravitational Wave, Corfu, Greece, Sep 2023.

**3.*** *Gravitational waves: principles of emission.*
Kavli-Villum Summer School on Gravitational Wave, Corfu, Greece, Sep 2023.

**2.*** *Introduction to gravitational waves (theory background).*
VIPER summer school on PTA GW astrophysics, Nashville TN, USA, (online), Jul 2022.

**1.** *Multiband gravitational-wave (astro)physics.*
Gravitational-wave Excellence through Alliance Training (GrEAT) PhD school, Birmingham, UK, Nov 2019.

## Posters at conferences:

**8.*** *Exploitation of gravitational-wave data.*
Milano-Bicocca Physics Department 25th anniversary, Milan, Italy, Sep 2024.

**7.** *PRECESSION: efficient black-hole binary evolution with python.*
Mathematics and Big Data showcase, University of Cambridge, Cambridge, UK, Apr 2016.

**6.*** *Numerical relativity group at University of Cambridge.*
Einstein's Legacy: celebrating 100 years of general relativity, London, UK, Nov 2015.

**5.*** *Formation and evolution of compact objects in relativity and modified gravity.*
5th DiRAC Science Day, Cambridge, UK, Sep 2015.

**4.** *New analytic solutions to binary black-hole dynamics: from spin precession to inspiral.*
Eurostrings 2015, Cambridge, UK, Mar 2015.

**3.*** *Analytic solutions to binary black-hole spin precession: recalling Kepler's two-body problem.*
Compact objects as astrophysical and gravitational probes, Leiden, The Netherlands, Jan 2015.
- **Best presentation award**.

**2.** *Efficient precession-averaged evolution of spinning black-hole binaries.*
RAS Specialist Meeting – Towards gravitational-wave astronomy: data analysis techniques and challenges, London, UK, Dec 2014.

**1.** *Morphologies and binary transfer: a new approach to the post-newtonian dynamics of precessing black-holes binaries.*
DPG Physics School "General Relativity @99", Bad Honnef, Germany, Sep 2014.

## Outreach talks:

**18.*** *Onde di gravità.*
MEETmeTonight European Researchers' Night, Milan, Italy, Sep 2025.

**17.*** *Buchi neri ed onde gravitazionali (aka: "Cosa racconto alla vostra prof. dopo una giornata di lavoro").*
Collegio San Carlo, Milan, Italy, Jun 2024.

**16.*** *Al confine fra astronomia e relatività.*
Finals of the Italian Physics Olympiad, Senigallia, Italy, Apr 2024.

**15.*** *Black holes on the way to merger.*
Cambridge University Astronomical Society, Cambridge, UK, Feb 2024.

**14.*** *Gravity with some data science in the blender.*
Datrix Tech Dissemination, Milan, Italy, (online), Oct 2023.

**13.*** *Hands-on astrophysics. Il ruolo del ricercatore.*
Italian Association of Physics Students (AISF), Milan, Italy, May 2023.

**12.*** *Gravity explorers.*
Istituto Svizzero, Milan, Italy, Mar 2023.

**11.*** *Onde gravitazionali, buchi neri e dove trovarli.*
Fondazione Sacro Cuore, Milan, Italy, Sep 2022.

**10.*** *Quando la gravità viaggia.*
Astronomiamo Association, Italy, (online), Mar 2022.

**9.*** *Hearing the Universe with gravitational waves.*
Bromsgrove Astronomical Society, Bromsgrove, UK, (online), Jun 2021.

**8.*** *Esplorare l'Universo con le onde gravitazionali.*
Settimana dell'Astronomia, Fondazione Lombardia per l'Ambiente, Italy, (online), Apr 2021.

**7.*** *Can black-hole binaries really make it?.*
University of Birmingham Astronomical Student Society, Birmingham, UK, (online), Mar 2021.

**6.*** *The (gravitational) sound of the Universe.*
Institute of Physics (IoP) Lecture Series, UK, (online), Dec 2020.

**5.*** *Looking at the Universe with gravity.*
Marling School visit, Birmingham, UK, Feb 2020.

**4.*** *Somewhere in between astronomy and relativity.*
Astronomy in the City, Birmingham, UK, Jan 2020.

**3.*** *Gravitational waves: somewhere between astronomy and relativity.*
Malvern Physics Olympics, Great Malvern, UK, Oct 2019.

**2.*** *Ride the wave (with gravity).*
Physics Day Experience, Birmingham, UK, Jun 2019.

**1.*** *Onde gravitazionali: ascoltare l'Universo (anziché solo vederlo?).*
Liceo Candia and Liceo Frassati, Seregno, Italy, Jan 2018.